Cultural Similarity and Team Performance: Evidence from Professional Doubles Tennis

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Abstract

In increasingly international and self-organized work environments, cultural similarity may be one dimension along which individuals self sort into collaborations. We examine relationship between cultural similarity and performance of self-organized teams using men's doubles tennis matches at the four Grand Slam tournaments. We measure cultural similarity using nationality, language, and linguistic proximity. Culturally similar teams are more likely to win matches. This association has a clear boundary: it disappears in tiebreaks, where teams are closely matched and outcomes are determined over few points. The association strengthens with professional experience but not with partners' experience gap. Finally, although cultural sorting in team formation is substantial, higher-ranked players are not systematically more likely to choose culturally similar partners, suggesting that the performance gain is not simply driven by sorting among stronger players.

1 Introduction

Many forms of work increasingly give individuals substantial discretion over whom they collaborate with, including entrepreneurial co-founders choosing business partners, researchers choosing co-authors, consultants assembling project teams, and professionals such as physicians, surgeons, and lawyers forming small working teams. As organizations and professional networks become increasingly international, these collaborations often bring together individuals from different national, linguistic, and cultural backgrounds. Among the many characteristics that may shape these choices, cultural similarity is one potentially important dimension of team formation process. Individuals may prefer culturally similar collaborators because shared language, norms, or backgrounds can facilitate communication, reduce coordination costs, and make it easier to anticipate one another's actions. This raises an important question: "When individuals have substantial discretion over whom they collaborate with, does cultural similarity between the resulting collaborators translate into better performance?"

The empirical evidence answering this question across corporate, academic, and sports suggests that teams sorted by cultural homophily often receive a performance penalty, though the exact boundaries remain heavily debated. In entrepreneurial and ventures, Gompers, Mukharlyamov, and Xuan (2016) find that ethnic homophily in self-selected entrepreneurial teams reduces investment success rates. In academics, Freeman and Huang (2015) document that culturally homogeneous research teams receiving fewer citations than their diverse counterparts. In sports, Békés and Ottaviano (2025) show that culturally similar players are more likely to pass the ball to one another within elite football teams; however, they note that such patterns do not necessarily translate into better overall team performance. Synthesizing these complex dynamics, a comprehensive meta-analysis by Stahl, Maznevski, Voigt, and Jonsen (2010) suggests that the ultimate effect of cultural diversity on team performance is not universal and depends heavily on the specific nature of the task environment, highlighting a critical need for research that can isolate these boundary conditions.

To empirically answer this need requires an alternative setting in which partner formation, real-time coordination, and subsequent team performance can all be observed simultaneously. Professional doubles tennis provides a unique real-world setting with

these exact desirable features. The workplace is international, where two individuals bilaterally form a partnership and operate under high-stakes economic incentives to maximize joint success in various competing conditions. Moreover, as individual contributions during play are directly observable, it limits concerns about unobserved effort and free-riding that frequently complicate team-performance measures in traditional corporate settings. Unlike academics—where unpublished research outputs or failed collaborations remain unobservable—the outcome of a doubles tennis match is always observable as a clean, binary win or loss. Similarly, unlike big team sports such as football—where players may have discretion over micro-level bilateral passing, but the entire team is ultimately assembled by a manager or coach—the collaboration in doubles tennis is entirely self-organized and organically formed.

In this paper, we leverage these unique institutional advantages to examine 1) the relationship between cultural similarity and team performance in a high-velocity working environment, 2) the boundary conditions of this relationship when communication windows are limited, and 3) the extent to which cultural similarity reflects systematic partner sorting in our setting.[1] By evaluating this setting, we are able to show how the benefits of cultural similarity depend on the immediate operational context and the specific time constraints under which teams must perform.

We collect data from all the men's doubles matches in Grand Slams between 2018 and 2025. Cultural similarity are measured along three complementary dimensions: shared nationality, shared official language, and language proximity. We first examine whether culturally similar teams are more likely to win their matches. We then use tiebreak outcomes to examine a theoretically important boundary condition: whether the association remains visible when the opportunity for teammates to coordinate is sharply compressed. Tiebreaks are played to decide a set once the score reaches 6–6. They are short and closely contested in which relatively few points determine the outcome and teammates have fewer opportunities to communicate and adjust during play.[2]

[1]Partnership formation in professional doubles tennis is inherently multidimensional. Players may sort on several characteristics, including playing ability and experience, strategic complementarity, and institutional or commercial connections such as shared representation by an agency or sponsorship arrangement. Our analysis abstracts from these multiple dimensions of partner selection and focuses on cultural similarity as one potentially consequential dimension of the resulting partnership.

[2]Much like a penalty shootout in football, a tennis tiebreak breaks a draw by shifting the match into a rapid-fire point-by-point format where losing a point on your own serve is the psychological equivalent

We find that culturally similar teams are 3.8 to 4.7 percentage points more likely to win matches, robust across all playing surfaces.[3] This association, however, disappears in tiebreaks, where the competing teams have already demonstrated closely matched performance, and the outcome is determined over a short sequence of points that provides relatively little time and opportunity for teammates to communicate, adjust, and coordinate. If culturally similar teams were simply stronger for some persistent reason unrelated to coordination, we would expect their advantage to remain visible in tiebreaks. Instead, the association is present when performance is aggregated over the entire match but not when the outcome is determined over a short, closely contested sequence. We interpret this contrast as evidence of a boundary condition on when cultural similarity can be associated with team performance, rather than as evidence of a universal performance advantage.

We then investigate how professional experience modulates this relationship by exploring two distinct human capital channels. First, a shared cultural background may function as a conduit that helps experienced teammates rapidly integrate their accumulated task-specific knowledge into coordinated action, implying that the association should strengthen with team average experience. Second, cultural similarity may act as a bridge for knowledge transfer between senior and junior teammates, suggesting the association should increase with the within-team experience gap. Our results support the former channel: the association between cultural similarity and match success is significantly amplified by team average experience, but is unaffected by the experience gap. This suggests that cultural similarity complements the total pool of accumulated professional human capital rather than serving as a tool to bridge internal status hierarchies.

Finally, we return to the endogenous team formation stage. We document the extent of cultural sorting in partnerships, examining whether the sorting is related to players' ability. Cultural similarity is substantially more common among observed partnerships (x%) than under random matching (x%), confirming that doubles teams are strongly sorted by nationality and language. However, this sorting is not more pronounced among

of missing a penalty kick, as the alternating sequence means a single miss often decides the entire set.

[3]Tennis Grand Slams are contested across three distinct structural profiles: grass at Wimbledon, clay at the French Open, and hardcourts at the Australian and US Opens. The surfaces may fundamentally alter game strategies and communication dynamics.

higher-ranked players, and a player's own ranking is, if anything, negatively associated with having a culturally similar partner. We therefore find no evidence for the specific selection explanation in which stronger players systematically choose culturally similar partners.[4] Nevertheless, this cannot rule out other forms of endogenous partner selection that we do not observe.

Our study relates to three strands of literature. First, it contributes to research on cultural diversity, or similarity, and organizational performance. Economic models emphasize that cultural diversity can generate informational and productivity benefits while also creating communication and coordination costs (Lazear, 1999). Organizational research similarly identifies both benefits and costs of cultural diversity, including greater creativity and knowledge variety but also conflict, weaker social integration, and coordination difficulties (Stahl et al., 2010; Harrison and Klein, 2007). Consistent with these competing forces, evidence on cultural composition and team performance is mixed and context dependent. Much of this literature examines diversity at the workforce or team level. We instead focus on cultural similarity within a small, highly interdependent dyad, where partners must coordinate continuously to produce a common outcome. In such settings, shared language and cultural background may directly facilitate communication and anticipation of a partner's actions.

Second, our study relates to the literature on endogenous team formation and professional team assembly. This literature emphasizes that teams may emerge through the choices of potential collaborators rather than solely through organizational assignment. Chen and Gong (2018), for example, show in a field experiment that self-selected teams outperform randomly assigned teams, with individuals primarily sorting according to pre-existing social networks. Other experimental studies show that the consequences of self-selection depend on skill complementarities and task characteristics (Büyükboyacı and Robbett, 2019; Fischer et al., 2023), while Foucart, Tan, and Zhao (2025) show how incentives can shape endogenous matching between high- and low-ability individuals. Evidence from professional settings likewise indicates that how temporary teams are assembled can affect coordination and performance (Tonellato et al., 2025). We complement this literature by focusing on cultural similarity as one dimension of endogenous

[4] Freeman and Huang (2015): if weaker scholars collaborate locally while stronger scholars can reach out internationally, it can cause culturally similar teams produce lower quality output. But this goes against our finding.

partner choice. We do not seek to model the full matching process: players may also sort on ability, experience, playing style, professional relationships, and commercial considerations. Rather, we examine whether cultural similarity is associated with performance conditional on key observable characteristics such as ranking and professional experience, and whether cultural similarity is systematically reflected in partnership formation.

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Third, our study contributes to research on cultural homophily, language, and coordination in collaboration. Language differences can impede information exchange and collaboration in multinational organizations (Marschan-Piekkari et al., 1999; Feely and Harzing, 2003), while linguistic similarity reduces communication frictions in economic interaction more broadly (Melitz and Toubal, 2014). Organizational research on shared mental models further emphasizes the role of common understandings of tasks and teammates in effective coordination (Cannon-Bowers et al., 1993; Klimoski and Mohammed, 1994; Mathieu et al., 2000). Most directly related, Békés and Ottaviano (2025) show that cultural homophily shapes pairwise collaboration among elite professional football players: culturally similar players are more likely to pass to one another within organizationally constituted teams. Their analysis establishes that culture affects who interacts with whom within an existing team, but does not examine whether such homophilous collaboration translates into aggregate team performance. We take the analysis one step further by asking whether cultural similarity between collaborators is associated with the objective performance of the resulting dyadic partnership, in a setting where continuous real-time coordination is central.

We make three distinct contributions. First, we connects organic selection directly to objective performance: We extend the literature on endogenous team formation by showing how bilateral cultural sorting translates into objective competitive outcomes, complementing studies like Békés and Ottaviano (2025) that map internal biases within forced team structures. Second, we identifies the temporal boundaries of cultural sorting: We show that the positive association between cultural similarity and performance is strictly bounded by the time window available for strategic communication and completely disappears under the acute time shocks of tiebreaks. Finally, we isolates the human capital pathways: We show that the homophily premium is heavily amplified by

the total pool of team expertise (team average experience) but is unaffected by internal status hierarchies (within-team experience gaps), supporting a coordination-based interpretation.

The remainder of the paper is organized as follows. Section 2 describes the data, measures of cultural similarity, and construction of the empirical sample. Section 3 develops the theoretical framework and formalizes our hypotheses. Section 4 presents the main empirical analysis, evaluating the baseline association between cultural similarity and performance, its robustness across playing surfaces as well its boundary condition, and the moderating pathways of professional experience. Section 5 returns to cultural sorting in the formation of partnerships and examines whether the sorting is related to players' ability. Section 6 concludes with a discussion of broader implications for both individual and organizational management.

2 Data

2.1 Empirical Setting and Sample

Our empirical setting consists of men's doubles matches from the four Grand Slam tournaments, the Australian Open, Roland Garros, Wimbledon, and the US Open, between 2018 and 2025. The dataset contains 1,864 completed Grand Slam doubles matches and 3,728 team-match observations.

Player information is collected by combining official tournament records from the Association of Tennis Professionals (ATP) and supplementary public sources. Nationality and language information are fully harmonized through a multi-stage validation procedure that resolves inconsistencies arising from spelling variants, accented characters, and historical nationality changes.

Table 1 reports the number of matches in each tournaments by year in Panel A and descriptive statistics at both individual and team level in Panel B.

2.2 Measuring Cultural Similarity

We construct three measures of cultural similarity from the literature. The first measure, *Same Nationality*, equals one if both teammates represent the same country and

Table 1: Observation Breakdown and Descriptive Statistics

Panel A: Matches by Tournament and Year (regression sample)

Tournament	2018	2019	2020	2021	2022	2023	2024	2025	**Total**
Australian Open	60	62	63	62	59	63	60	61	**490**
Roland Garros	63	60	62	62	62	61	58	63	**491**
Wimbledon	61	61	0	60	59	61	61	62	**425**
US Open	63	60	31	59	62	61	63	59	**458**
Total	**247**	**243**	**156**	**243**	**242**	**246**	**242**	**245**	**1,864**

Note: In 2020, due to Covid, Wimbledon did not take place and US Open 2020 halved the draw of men's doubles. From the initial dataset of 1,921 Grand Slam doubles matches, we exclude retirements and walkovers because these outcomes do not reflect normal competitive performance. We further remove matches with incomplete doubles ranking information that cannot be reliably recovered, resulting in a final sample of 1,864 matches.

Panel B: Descriptive Statistics

	Individual level (N=7,456)		Team level (N=3,728)	
Variable	Mean	SD	Mean	SD
Doubles ranking	126.02	210.46	126.02	183.51
Years since turning pro	12.59	5.36	11.93	5.17
Age at tournament	31.57	5.19	31.60	4.36
Top-100 singles player (share)		–		8.1%

Note: Team-level statistics are the two-teammate average used directly in the regressions (*rank_mean*, *exp_mean*, *age_mean*). Years since turning pro is imputed to 1 for 216 team-obs whose raw tenure is ≤ 0 (rookies who turned pro in or just after the tournament year).

zero otherwise.[5] The second measure, *Same Language*, equals one if both teammates share an official national language and zero otherwise. We construct this measure from CEPII's Gravity database (Conte et al. 2022), which records, for each pair of countries, whether they share an official language.[6]The third measure, *Linguistic Proximity*, is a continuous index on $[0, 1]$ built based on the Ethnologue's language-tree classification from the data repository in Melitz and Toubal (2014).[7] This continuous measure distinguishes teammates who speak closely related but distinct languages (e.g. Italian and Spanish) from teammates whose languages are totally unrelated.[8] It is also one of the

[5]ATP players compete under a single playing nationality, so each player contributes exactly one country code to this measure and no dual-nationality assignment rule is needed.

[6]The official language is defined at the country level rather than at the individual level, that is, a player who is bilingual, or whose own mother tongue differs from his playing nationality's majority language, is assigned the value implied by that nationality.

[7]An alternative binary construction of linguistic proximity (CEPII's *comlang_ethno*, one if both teammates share a widely-spoken ethnic language and zero otherwise, defined as a language spoken as a mother tongue by at least 9% of the population in both countries) and two further continuous constructions from Pellegrino et al. (2024) are reported as robustness checks in Appendix A.

[8]Melitz and Toubal (2014) score each pair of *languages* on a coarse ordinal scale according to how many levels of the Ethnologue family tree they share (e.g. two languages in the same branch but a

same measures used in Békés and Ottaviano (2025) to characterize linguistic similarity between teammates in football teams.

These measures reflect increasingly broad notions of cultural similarity. First, shared nationality captures a broad form of common cultural background, including familiarity with norms, conventions, social references, and behavioral expectations. Second, shared language provides a more direct channel through which partners can communicate intentions, tactical information, and feedback, potentially reducing the costs of developing compatible expectations. Third, linguistic proximity captures more gradual forms of similarity between languages and may reduce communication and interpretation frictions even when partners do not share an official language.

Table 2 reports the share of teams by each measure. In our sample, 42.2% (821 teams) of teams consist of players sharing the same nationality, 56.1% (1,091 teams) share an official language, and the mean of the continuous measure linguistic proximity is 0.651.

Table 2: Shares of the Three Cultural Similarity Measures

Measure of Cultural Similarity	Share or Mean	No. of Distinct Partnerships
Same nationality	42.2%	821
Same language	56.1%	1,091
Ling. proximity	0.651	1,945

Note: Computed over the 1,945 distinct tournament×partnership pairings with complete ranking data (510 at the Australian Open, 511 at Roland Garros, 447 at Wimbledon, 477 at the US Open). "Teams" is the absolute number of the 1,945 partnerships satisfying each binary criterion; Linguistic proximity is a continuous $[0, 1]$.

3 Theory and Hypotheses

3.1 Cultural Similarity and Team Performance

A central idea in the organizational literature on team coordination is that team members develop mental models of the task and of one another that guide how they interpret

different sub-branch score 0.5), and the resulting *country*-pair value is a population-weighted average across each country's principal languages. For example, Belgium is absent from their table; we then construct its value against any other country as the population-weighted average of the Netherlands' and France's own values against that country, using Belgium's approximate Dutch/French population shares (59%/41%) as weights, the same way Melitz and Toubal (2014) themselves apply to other multi-language countries such as Switzerland. This gives Belgium a linguistic-proximity value of 0.56 against France ($0.59 \times 0.25 + 0.41 \times 1.00$: Dutch and French sit in different branches of the same language family, while French scores trivially 1.00 against itself) and 0.69 against the Netherlands ($0.59 \times 1.00 + 0.41 \times 0.25$).

situations and coordinate their actions (Cannon-Bowers et al., 1993; Klimoski and Mohammed, 1994). When these mental models are sufficiently compatible, team members can anticipate one another's actions, divide responsibilities, and respond to changing circumstances without having to explicitly communicate every decision. More broadly, research on team cognition emphasizes that effective joint performance depends not only on the capabilities of individual members but also on their ability to develop compatible expectations about the task and about one another.

Cultural similarity may facilitate this process by providing common ground for communication and interpretation. In our three measures of cultural similarity, shared nationality may capture familiarity with common norms, conventions, social references, and behavioral expectations; shared language can directly reduce communication and interpretation costs by facilitating the exchange of intentions, tactical information, and feedback; linguistic proximity may provide some of these benefits even when partners do not share an official language, by reducing the difficulty of understanding one another. These forms of similarity may make it easier for partners to anticipate one another and coordinate their individual actions.

This logic is particularly relevant to doubles tennis. Partners jointly produce the match outcome and must continuously coordinate positioning, shot selection, court coverage, and tactical responses. Many adjustments occur rapidly and repeatedly, leaving limited scope for every action to be specified through explicit communication. The ability to anticipate a partner's likely action and respond appropriately is therefore central to effective joint performance. Cultural similarity may facilitate this process by lowering communication and interpretation frictions and making it easier for partners to develop compatible expectations about play. We therefore propose:

Hypothesis 1. *Greater cultural similarity within teams is positively associated with the probability of winning a match.*

3.2 Tiebreaks as a Boundary Condition

The coordination benefits associated with cultural similarity are not necessarily equally observable in every part of a match. Coordination is a process that develops through repeated interaction: teammates communicate, adjust their behavior, learn one an-

other's tendencies, and build shared expectations. While a full match provides a series of opportunities for this process to operate, a tiebreak imposes a different temporal and competitive environment in which the opportunity for any coordination advantage to accumulate is substantially reduced.[9] First, tiebreaks are short relative to the match as a whole, leaving less time for a coordination advantage to accumulate. Second, a tiebreak is reached only when the two teams have performed sufficiently similarly to reach a closely contested stage of the set. Its outcome is therefore determined in a setting where relatively small differences in performance can be decisive and where point-level randomness is likely to play a larger role.

These features make the tiebreak a particularly informative boundary condition. If cultural similarity improves team performance partly by facilitating communication in adjusting tactics and coordinating subsequent play, its advantage should be less apparent in a short, closely contested sequence in which there are fewer opportunities for communication and greater scope for point-level randomness. This leads to our second hypothesis:

Hypothesis 2. *The positive association between cultural similarity and team performance is weaker when the opportunity for teammates to coordinate is limited, such as in short, closely contested episodes.*

3.3 Professional Experience as a Moderating Factor

Professional experience provides a further way to examine whether the observed relationship is consistent with a coordination-based interpretation. Experience may affect how effectively players are able to use the communication and interpretive resources associated with cultural similarity. More experienced professionals have accumulated greater exposure to competitive situations and may be better able to translate shared

[9] Two distinct tiebreak formats exist in professional tennis. A *standard* tiebreak, played to decide a set once the game score reaches 6–6, is won by the first team to reach 7 points with a margin of at least 2 (so it typically lasts 7–10 points in practice); this format is used in every set of a match, and, at some tournaments, in the final set as well. A *match* or "super" tiebreak, played to 10 points instead of 7, is used at some tournaments as a substitute for a full final set. In our sample, no genuine 10-point match tiebreak is ever used to decide the last set at any of the four Grand Slams. Wimbledon is the only tournament with a non-standard deciding-set format: in 15 of our sample's matches (7 in 2018, 3 in 2019, 5 in 2021), the last set was completed through extended advantage-set play with no breaker at all, with final scores such as 16–14 and 22–20, consistent with Wimbledon's pre-2019 rule of no tiebreak in the deciding set and, for the 2019 and 2021 cases, with its subsequent 12-12-breaker rule simply never being triggered because the set ended before reaching 12-12. At the Australian Open, Roland Garros, and US Open, and in the remaining 79 Wimbledon matches with a completed deciding set, the last set is completed by the same standard 6–6 seven-point breaker.

linguistic or cultural understanding into rapid anticipation and coordinated action. Following this logic, the association between cultural similarity and performance should become stronger as the partners have accumulated more professional experience.

Professional experience may also matter through a different channel. Cultural similarity may be particularly valuable when partners differ substantially in professional experience. A larger experience gap may increase the need for knowledge transfer between teammates, as the less experienced player may benefit from the more experienced player's tactical knowledge, situational judgment, and understanding of competitive play. When partners share cultural or linguistic common ground, this knowledge may be easier to communicate and interpret. If cultural similarity facilitates such knowledge transfer, its positive association with team performance should be stronger in partnerships with larger within-team experience gaps.

We therefore examine two complementary dimensions of professional experience. First, we consider the team's average professional experience to assess whether cultural similarity is more strongly associated with performance among more experienced partnerships. Second, we examine the within-team experience gap to assess whether cultural similarity is particularly relevant when partners differ substantially in professional experience. These analyses are intended as evidence bearing on the proposed coordination interpretation rather than as tests of additional causal mechanisms.

4 Empirical Analysis

4.1 Match Performance

We test Hypothesis 1 by estimating the association between cultural similarity of the two players within a team and the probability of winning a doubles match. Specifically, we estimate a Logit model

$$\Pr(W_{ijt} = 1) = f(\beta C_{ijt} + \gamma X_{ijt} + \delta_{ty} + \lambda_r), \tag{1}$$

where W_{ijt} equals one if team i wins match j in tournament-year t; C_{ijt} denotes one of our three measures of cultural similarity: same nationality, same language, and linguistic proximity; X_{ijt} includes three sets of control variables. First, observable own

team and opponent team qualities, measured using the average ATP individual doubles ranking of each players.[10] Lower ranking values indicate stronger players. Second, an indicator for whether either teammate in own team is ranked inside the ATP Top 100 in singles, capturing exceptional individual talent that may not be reflected fully in doubles rankings. Third, professional experience is measured as the average number of years since the two teammates turned professional. To allow for diminishing returns, we also include its squared term.

Throughout the analysis we include tournament-by-year fixed effects δ_{ty} and round fixed effects λ_r to absorb common variation across tournaments and stages of competition. Standard errors are clustered at the match level to account for interdependent team-match observations.

Table 3: Match Win — Average Marginal Effects (AME) from Logit

	(1) Same Nationality	(2) Same Language	(3) Ling. Proximity
Cultural similarity	0.0361**	0.0420***	0.0501**
	(0.0164)	(0.0157)	(0.0202)
Own avg. doubles ranking	−0.0007***	−0.0007***	−0.0007***
	(0.0001)	(0.0001)	(0.0001)
Opponent avg. doubles ranking	0.0007***	0.0007***	0.0007***
	(0.0001)	(0.0001)	(0.0001)
Top-100 singles player (0,1)	−0.0042	−0.0032	−0.0031
	(0.0315)	(0.0315)	(0.0314)
Team avg. professional years	0.0044***	0.0042**	0.0043***
	(0.0017)	(0.0017)	(0.0017)
Team avg. professional years^2	−0.0008***	−0.0008***	−0.0008***
	(0.0002)	(0.0002)	(0.0002)
Tournament × year FE	Yes	Yes	Yes
Match round FE	Yes	Yes	Yes
Pseudo-R^2	0.096	0.097	0.096
N	3,728	3,728	3,728

*** p<0.01, ** p<0.05, * p<0.10. Standard errors in parentheses, clustered by match. In Column (1), cultural similarity is equal to 1 if teammates have the same nationality, and zero otherwise. In Column (2), cultural similarity is equal to 1 if teammates speak the same official language, and zero otherwise. In Column (3), cultural similarity between teammates is measured by linguistic proximity on a 0-1 continuous scale. Unit of observation is team-match (2 team obs. per match); Professional years is demeaned at the sample mean (11.93 years)

[10] The ATP does not publish a persistent, tournament-by-tournament doubles ranking at the pair level: the doubles rankings are at the individual level. The only ATP ranking computed at the pair level is the Doubles Race, used exclusively to seed qualification for the 8 spots in the year-end ATP Finals; it resets every calendar year and is defined only for pairs that have accumulated enough joint results together in that season. Hence, it does not exist for a large share of our sample, since doubles partnerships change frequently across tournaments, and many Grand Slam pairings are new or infrequent. Each player's own individual double ranking, by contrast, is continuously available regardless of current partner, so we use the two teammates' average individual ranking as the team-quality measure.

Table 3 reports average marginal effects (AMEs) estimated in logit models. Across all three specifications, culturally similar teams are significantly more likely to win Grand Slam matches. The estimated effects range from 3.6 to 5.0 percentage points and are largest for linguistic proximity. Team quality are the dominant predictor of success: stronger own doubles rankings significantly increase the probability of winning, while stronger opponent rankings significantly decrease the probability of winning. Whether having a top 100 singles player in the team does not seem to matter for the match outcome. More experienced teams perform better with diminishing returns.

DOUBLE CHECK THE Alternative specification: randomly assign one team per match as the focal team to be kept in the regression analysis (N=1,864). Coefficients are of similar magnitude and slighter weaker significance. Repeated over 10 random seeds (Appendix Table 15): the same-nationality AME averages 0.0466 (range 0.0161–0.0812, significant at the 5% level in 5 of 10 draws), same-language averages 0.0477 (range 0.0215–0.0847, significant in 6 of 10 draws), and, for linguistic proximity averages 0.0656 (range 0.0236–0.0944, significant in 7 of 10 draws).

These findings provide consistent evidence that cultural similarity is positively associated with team performance after accounting for observable differences in own and opponent quality, experience, and tournament characteristics.

4.2 Robustness check across Competing Environments

An important question is whether the association between cultural similarity and team performance persists across different competitive environments or depends on the conditions under which teams perform. In this subsection we perform a robustness check of the observed association across different competing environments.

One of the main differences across Grand Slam tournaments is playing surface. The surface affects ball speed, bounce, and patterns of play, potentially changing the tactical demands placed on partners. We therefore examine whether the association between cultural similarity and match outcomes varies across these environments. We pool the US Open and Australian Open as the hard-court baseline and compare them with the French Open on clay and Wimbledon on grass. Because each surface is represented by a specific set of tournaments, surface characteristics cannot be fully disentangled

from other tournament-specific features. We therefore interpret this analysis as a test of robustness across tournament-surface environments, rather than as an isolated test of the effect of court surface. Similar associations across these environments would suggest that the main finding is not driven solely by one particular competitive setting, whereas substantial differences would indicate that the relationship between cultural similarity and performance may be context dependent.

Table 4 reports the results. The estimated coefficients of cultural similarity remain remarkably stable in sign and magnitude across surfaces. The interaction estimates provide little evidence that the association between cultural similarity and match performance differs systematically across playing environments. The estimated association on grass is not meaningfully different from that on clay and hard courts combined, nor is the association on clay meaningfully different from that on grass and hard courts combined. This consistency suggests that the main finding is not concentrated in a particular tournament-surface environment. However, because surface is tied to tournament identity in our sample, these comparisons cannot disentangle surface-specific conditions from other tournament characteristics. Overall, the results support the robustness of the cultural-similarity association across the competitive environments observed in Grand Slam doubles tennis.[11]

Table 4: Results across playing surfaces

	(1) Same Nationality		(2) Same Language		(3) Ling. Proximity	
	(a) grass	(b) clay	(a) grass	(b) clay	(a) grass	(b) clay
Cultural similarity (C)	0.0395**	0.0351*	0.0392**	0.0442**	0.0541**	0.0486**
	(0.0183)	(0.0189)	(0.0176)	(0.0182)	(0.0224)	(0.0234)
Surface	0.0097	−0.0017	−0.0043	0.0052	0.0158	−0.0028
	(0.0149)	(0.0142)	(0.0208)	(0.0189)	(0.0308)	(0.0285)
C × Surface	−0.0169	0.0023	0.0112	−0.0094	−0.0203	0.0034
	(0.0376)	(0.0347)	(0.0364)	(0.0340)	(0.0474)	(0.0438)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Round FE	Yes	Yes	Yes	Yes	Yes	Yes
Tournament × year FE	No	No	No	No	No	No
Pseudo-R^2	0.096	0.096	0.097	0.097	0.096	0.096
N	3,728		3,728		3,728	

*** p<0.01, ** p<0.05, * p<0.10. Standard errors in parentheses, clustered by match. We pool the US Open and Australian Open as the hard-court baseline and compare them with the French Open on clay and Wimbledon on grass. Spec (a): grass vs. rest (hardcourt+clay pooled). Spec (b): clay vs. rest (hardcourt+grass pooled). Round FE only; Same controls as in Table 3.

[11] As a robustness check, restoring tournament×year FE and estimating C×grass/C×clay jointly (hardcourt as baseline, no separate surface main effect) leaves this conclusion unchanged: both C×grass and C×clay interactions remain small and insignificant (p>0.65) in that specification.

4.3 Testing Tiebreaks as a Boundary Condition

The match-level results establish a positive association between cultural similarity and team performance. If cultural similarity is associated with performance because shared cultural and communication backgrounds facilitate coordination between teammates, its performance relevance should depend partly on whether teammates have sufficient opportunity to communicate and adjust accordingly. The longer the task, the more opportunities teammates have to develop and apply such coordination. Therefore, we next ask whether this association persists when the opportunity for team coordination is substantially compressed. This provides a boundary condition for the interpretation of the main results.

A tennis tiebreak is a short and highly compressed sequence in which relatively few points determine the outcome.It is played after the two teams have already demonstrated closely matched performance within the match. The outcome is therefore determined over a much shorter sequence than the match as a whole, leaving less time and opportunity for communication-related coordination advantages to accumulate. At the same time, point-level variation is likely to account for a larger share of the final outcome. Thus, even if cultural similarity is associated with better coordination over the course of a match, its potential performance benefits may be too small to become visible when the competing teams are sufficiently close in performance and the outcome is determined over only a small number of points.

We estimate the same specification used for the match-level analysis, replacing the match-win outcome with an indicator for winning the tiebreak. Table 4 reports the results. Across all three measures of cultural similarity, the estimated associations are substantially smaller than those observed for match outcomes and are not statistically distinguishable from zero. The absence of a detectable association is consistent across shared nationality, shared official language, and linguistic proximity.

The contrast between the two outcomes is informative. If culturally similar teams were simply stronger for some persistent reason unrelated to coordination, there would be less reason to expect their advantage to disappear precisely when the outcome is compressed into a short sequence. Instead, the association is visible when performance is aggregated over the course of the match but not when the outcome is determined over

a brief, closely contested episode. We interpret this pattern as a boundary condition on the performance relevance of cultural similarity: potential coordination benefits may require sufficient time and opportunity to emerge and accumulate, and may therefore be difficult to detect when teams are closely matched and the task provides too little scope for teammates to communicate and adjust.

We estimate the same empirical specification in Equation (1) using tiebreak results as the outcome variable.

Table 5: Tiebreak Win — Average Marginal Effects (AME) from Logit

	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Cultural similarity	0.0308	0.0087	0.0388
	(0.0224)	(0.0218)	(0.0277)
Own avg. doubles ranking	−0.0003***	−0.0003***	−0.0003***
	(0.0001)	(0.0001)	(0.0001)
Opponent avg. doubles ranking	0.0003***	0.0003***	0.0003***
	(0.0001)	(0.0001)	(0.0001)
Top-100 singles player (0,1)	−0.0675	−0.0646	−0.0672
	(0.0428)	(0.0427)	(0.0428)
Team avg. professional years	0.0027	0.0023	0.0026
	(0.0022)	(0.0021)	(0.0022)
Team avg. professional years^2	0.0000	0.0001	0.0000
	(0.0003)	(0.0003)	(0.0003)
Tournament × year FE	Yes	Yes	Yes
Round FE	Yes	Yes	Yes
Pseudo-R^2	0.013	0.013	0.013
N	2,354	2,354	2,354

*** p<0.01, ** p<0.05, * p<0.10. Standard errors in parentheses, clustered by match. Unit of observation is team-tiebreak (2 team obs. per tiebreak); standard 7-point tiebreaks only. Same controls and FE as Table 3.

Table 5 reports the results. None of the three measures of cultural similarity is significantly associated with the probability of winning a tiebreak. The ranking controls remain highly significant, indicating that the specification continues to capture underlying differences in team quality.

If culturally similar teams were simply stronger for some persistent reason, we would have some reason to expect that advantage to remain visible in tiebreaks.

Taken together, these findings suggest that the association between cultural similarity and team performance has an important condition: it is more likely to emerge when teammates have sufficient opportunity to coordinate over the course of a task. The positive association with match outcomes, combined with its absence in tiebreaks,

is consistent with cultural similarity mattering through coordination processes that can accumulate over time, rather than providing an advantage that operates equally in every episode of team performance. If culturally similar teams were simply stronger for some persistent reason, we would expect that advantage to remain visible in tiebreaks. Instead, the association is present when performance is aggregated over the match but not when the outcome is determined over a short, closely contested sequence. Tiebreaks therefore provide a particularly stringent boundary condition: when two teams are already performing very similarly, the potential performance benefit of cultural similarity may be too small to become visible over a short sequence, in which teammates have limited time and opportunity to communicate, adjust to one another, and translate coordination advantages into the final outcome.

4.4 Professional Experience as a Moderating Factor

In this section we examine the roles of players' professional experience which could shed light on potential mechanisms. Since our data do not directly observe communication or coordination processes, these analyses should be interpreted as mechanism-related heterogeneity tests rather than direct tests of the mechanisms themselves.

First, players with more years of professional experience are likely to have accumulated greater task-specific knowledge and a better understanding of the demands of high-level competition. When teammates share a language or broader cultural background, communication may make it easier to convey and integrate this experience into joint decision-making, allowing culturally similar teams to translate their individual professional knowledge into coordinated action more effectively. If this channel is important, the association between cultural similarity and performance should be stronger among teams whose members have greater professional experience.

Second, cultural similarity may facilitate knowledge transfer when teammates differ substantially in professional experience. A shared language and cultural background may reduce communication barriers and make it easier for a more experienced player to communicate task-relevant knowledge to a less experienced partner. Under this interpretation, the association between cultural similarity and performance should become stronger as the experience gap within the team increases.

4.4.1 Average Professional Experience

We first test whether the association between cultural similarity and match performance varies with team average experience. For each team, we calculate the average number of years since the two players turned professional. Notice that this is not the years the two players have been working together in the same team, but the average individual professional experience of the two players.[12] We interact this measure with cultural-similarity while retaining the full set of controls and fixed effects from the baseline specification in Equation (1). That is, we estimate the following model

$$\Pr(\text{Win}) = f(\beta_1 C + \beta_2 \text{AvgTeamExp} + \beta_3 (C \times \text{AvgTeamExp}) + \gamma X + \delta_{ty} + \lambda_r) \quad (2)$$

Table 6 reports the results. The interaction between cultural similarity and team average experience is positive for all three measures. For same nationality, however, the interaction is very small and statistically insignificant (0.0008, p=0.809). For shared language, the interaction is positive and statistically significant (0.0070, p=0.029). For linguistic proximity, the interaction is positive but no statistically significant (0.0053, p=0.207)

Table 6: Team Average Experience

	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Cultural similarity (C)	0.0361**	0.0394**	0.0473**
	(0.0164)	(0.0157)	(0.0203)
C × exp_mean_dm	0.0008	0.0070**	0.0053
	(0.0033)	(0.0032)	(0.0042)
exp_mean_dm (main)	0.0041*	0.0002	0.0007
	(0.0023)	(0.0025)	(0.0033)
Controls	Yes	Yes	Yes
Tournament × year FE	Yes	Yes	Yes
Round FE	Yes	Yes	Yes
Pseudo-R^2	0.096	0.098	0.097
N	3,728	3,728	3,728

*** p<0.01, ** p<0.05, * p<0.10. Standard errors in parentheses, clustered by match. *exp_mean_dm* is demeaned at the sample mean (11.93 years, SD 5.17), so the Culture (C) row is the AME at mean team experience.

[12] In tennis, partnership can change relatively often. Players may team up with different partners in different tournaments. Calculate how many teams and the share in total have repeated partnership during the sample period and how many unique team appearance.

The result is robust to allowing the interaction to vary nonlinearly with experience. When we additionally include the interaction between cultural similarity and squared demeaned experience, the quadratic interactions remain small and statistically insignificant for all three measures.

This pattern is consistent with a mechanism based on accumulated professional experience and the development of effective coordination. More experienced players may be more able to anticipate teammates' actions and establish efficient communication routines. Cultural and linguistic similarity may make these coordination processes more effective, so that their association with performance becomes stronger among more experienced teams. The fact that the amplification result is concentrated in shared official language, and absent for both shared nationality and the continuously-measured linguistic proximity, is consistent with a communication-based interpretation tied specifically to official/formal language use, rather than to cultural similarity or graded linguistic closeness more broadly.

Importantly, our measure captures players' individual professional experience, rather than the number of matches a particular pair has played together. We therefore do not directly observe the development of pair-specific shared mental models or communication routines. The evidence should consequently be interpreted as consistent with, rather than establishing, a process in which professional experience strengthens the performance relevance of cultural and linguistic similarity.

4.4.2 Experience Gaps within the Team

On the other hand, shared language becomes especially valuable when teammates differ substantially in expertise because communication and knowledge transfer become more demanding. Large experience gaps imply asymmetric knowledge, asymmetric decision authority, coaching/mentoring, and more need to transfer tacit knowledge. That is exactly where shared language may matter most. The experience-gap analysis asks whether cultural similarity is particularly valuable when teammates differ in professional experience, conditional on the team's overall competitive strength and average professional experience. That is, among teams with comparable overall strength and comparable average professional experience, does cultural similarity become particu-

larly valuable when the two teammates differ more in professional experience?

The estimating equation becomes

$$\Pr(\text{Win}) = f(\beta_1 C + \beta_2 \text{TeamExpGap} + \beta_3 (C \times \text{TeamExpGap}) + \gamma X + \delta_{ty} + \lambda_r) \quad (3)$$

The second mechanism makes a different prediction. Cultural similarity may matter particularly when teammates have different levels of professional experience. An experienced player may possess greater knowledge of the demands of elite doubles competition, while an inexperienced teammate may need to learn unfamiliar routines and expectations. Under this knowledge-transfer interpretation, a common cultural and linguistic background could facilitate communication between teammates with different levels of experience. The value of cultural similarity should therefore increase with the experience gap within the team, rather than with the average experience of the two players.

We test this prediction using the absolute difference between the two teammates' years since turning professional, ExperienceGap=. We interact this measure with each cultural-similarity measure and additionally control for team average experience and its quadratic term. The experience gap is demeaned at its sample mean of 5.50 years, so the cultural-similarity coefficient is evaluated at the average experience gap rather than at zero. The models otherwise retain the specification used in the main analysis.

The results provide no evidence that cultural similarity becomes more strongly associated with performance as the experience gap increases. The interaction is positive for all three measures—0.0032 for same nationality (p=0.344), 0.0040 for same language (p=0.214), and 0.0031 for linguistic proximity (p=0.447)—but none is statistically significant. Allowing for a quadratic relationship between the experience gap and cultural similarity produces the same conclusion: the quadratic interactions are small and statistically insignificant, while the linear interactions remain insignificant.

The results therefore distinguish the two potential mechanisms. We find evidence that the association between cultural similarity and performance becomes stronger among more experienced teams, particularly for language-related measures, but no evidence supporting the specific prediction that cultural similarity is especially valuable as a bridge across unequal experience levels. The findings are consequently more consistent with an interpretation based on accumulated professional experience and coordination.

Table 7: Within-Team Experience Gap

	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Culture (C) – mean gap	0.0356**	0.0408***	0.0473**
	(0.0164)	(0.0157)	(0.0203)
C × exp_gap_dm	0.0032	0.0040	0.0031
	(0.0034)	(0.0032)	(0.0041)
exp_gap_dm (main)	−0.0036*	−0.0045*	−0.0043
	(0.0021)	(0.0025)	(0.0031)
Controls (Table 3 set)	Yes	Yes	Yes
Tournament × year FE	Yes	Yes	Yes
Round FE	Yes	Yes	Yes
Pseudo-R^2	0.097	0.097	0.097
N	3,728	3,728	3,728

*** p<0.01, ** p<0.05, * p<0.10. Standard errors in parentheses, clustered by match. *exp_gap_dm* $= |\exp_i - \exp_j|$, demeaned at the sample mean (5.50 years, SD 4.92); individual tenure is imputed to 1 year for rookies. Controls include team-average *exp_mean_dm* and its square, plus *exp_gap_dm* itself. A robustness spec adding C × exp_gap_dm^2 leaves the conclusion unchanged (quadratic term and linear term both insignificant throughout, p>0.29).

This distinction is important for interpreting the mechanism. We do not find evidence that cultural similarity primarily helps an experienced player transmit knowledge to a less experienced teammate. Instead, the results point toward a setting in which the value of cultural and linguistic compatibility is greater among teams whose members have accumulated more professional experience. While this pattern is consistent with the gradual development of shared expectations and coordination routines, the available data do not allow us to observe those processes directly.

5 Team Formation and Cultural Sorting

Partnership formation in professional doubles tennis is inherently multidimensional. Players may sort according to ability, experience, playing style, professional relationships, or commercial connections, among other considerations. We focus here on one dimension of this broader matching process: cultural similarity. An important feature of doubles tennis is that teams are formed bilaterally between players rather than assigned exogenously. This makes partnership formation relevant for interpreting the performance associations documented above. We therefore examine partnership formation separately from the performance analysis.

The purpose of this section is not to eliminate selection concerns but to document how cultural sorting operates in the doubles environment and whether it is particularly pronounced among stronger players which could directly lead to the observed positive association between cultural similarity and performance. We first compare observed cultural similarity in partnerships with the corresponding random-matching benchmark. Subsequently, we examine whether a player's own doubles ranking predicts the cultural similarity of their partner. Together, these analyses provide a more transparent picture of the selection environment in which the performance associations arise.

5.1 Cultural Sorting in Partnership Formation

We compare the observed prevalence of cultural similarity with the rate expected under random matching. The benchmark is built in two steps. First, for each tournament-year separately (e.g., Wimbledon 2018), we take the players in that field and compute the share of all possible player pairs that would satisfy the relevant cultural criterion. This gives the rate we would expect if players in that tournament had chosen their partners at random, and it accounts for the nationality and language composition of each field. Second, we pool these tournament-year benchmarks into the single figure reported in Table 8 by taking their average, weighted by the number of actual partnerships in each tournament-year. Both rows therefore weight each tournament-year in the same way (by its number of partnerships), so the gap between them reflects partner choice rather than differences in how tournaments are averaged. For the top-100 and top-50 columns, both the player pool and the weights are restricted to players in the relevant ranking bracket.

Table 8: Cultural Sorting in Partnership Formation: Actual vs. Random-Matching Benchmark

Measure		All players (N=1,945)	Both in top-100 (N=1,083)	Both in top-50 (N=606)
Same nationality	Actual	42.2%	31.9%	35.0%
	Random benchmark	5.1%	3.9%	3.9%
Same language	Actual	56.1%	48.2%	54.5%
	Random benchmark	16.2%	15.3%	15.1%
Ling. proximity (mean)	Actual	0.651	0.593	0.621
	Random benchmark	0.361	0.357	0.370

Note: There are 1,945 distinct partnership per tournament between 2018–2025. A partnership is counted once within a tournament, even if the same pair advances through several rounds. Random benchmark is the closed-form expected rate under random pairing within each tournament-year's actual field, averaged across tournament-years with weights equal to the number of partnerships in each.

The results in Table 8 show substantial cultural sorting. Among all partnerships, 42.2% consist of players of the same nationality, compared with 5.1% under random matching. Similarly, 56.1% of partnerships share an official language, compared with 16.2% under random matching. Mean linguistic proximity is 0.651 among observed partnerships, compared with 0.361 under random matching. Thus, doubles partnerships are strongly assortative along all three dimensions of cultural similarity.

The sorting does not appear to be concentrated among the strongest players. Among partnerships in which both players are ranked in the top 100, 31.9% are same-nationality partnerships, compared with 3.9% under random matching. Among partnerships with both players in the top 50, the corresponding figures are 35.0% and 3.9%. Similar pattern is evident for shared language and linguistic proximity.

This fact is important for interpreting the performance results. Cultural similarity is clearly not randomly distributed across partnerships: players who form teams are much more culturally similar than would be expected from the composition of the tournament field alone. The match-performance estimates should therefore not be interpreted as if cultural similarity were randomly assigned. Rather, they describe an association between the cultural composition of an exiting partnership and its subsequent performance. However, these comparisons provide no evidence that the cultural sorting observed is disproportionately driven by the strongest players.

5.2 Player Ranking and Partner Selection

We next examine the relationship between a player's own ranking and the cultural similarity of the partnership more directly. For each partnership, we treat each player in turn as the focal player, generating two player-level observations per partnership. This produces 3,890 player-partnership observations from the 1,945 partnerships. Because the same partnership contributes two observations, standard errors are clustered at the partnership level.

We regress each cultural-similarity measure on the focal player's own doubles ranking as the baseline specification, including tournament-by-year fixed effects. Since the cultural composition of the player pool differs across tournament-years, a player's potential choice set of a culturally similar partner may vary mechanically with the set of players entering the event. We therefore extend the baseline specification to account for the cultural composition in the field, corresponded to each cultural measure, of the tournament field: for each focal player, we compute the share of *other* players in that tournament-year sharing the same nationality and language with the focal player, as well as the analogous mean pairwise value of linguistic proximity against every other player in the field. Formally, for a focal player i entering a tournament-year field F_t of N total entrants,

$$\text{FieldNatComposition}_{it} = \frac{1}{N_t - 1} \sum_{\substack{j \in F_t \\ j \neq i}} \mathbf{1}\{\text{Nationality}_j = \text{Nationality}_i\};$$

$$\text{FieldLangComposition}_{it} = \frac{1}{N_t - 1} \sum_{\substack{j \in F_t \\ j \neq i}} \mathbf{1}\{\text{Language}_j = \text{Language}_i\};$$

$$\text{FieldLingProxComposition}_{it} = \frac{1}{N_t - 1} \sum_{\substack{j \in F_t \\ j \neq i}} \text{LingProx}_{ij}$$

As an illustration, at Roland Garros 2019, an Italian player faced a field of 4 (Berrettini, Sonego, Seppi, Cecchinato) Italians (3 others besides himself) among 127 other entrants, giving FieldNatComposition = FieldLangComposition = $3/127 = 0.024$, since no other country speaks Italian as an official language. The corresponding linguistic proximity score is substantially higher at 0.344, driven by players speaking related languages.

For instance, 20 French players in the same tournament – each sharing an Ethnologue Romance branch with Italian ($Ling_Prox = 0.5$) – contributed 10.0 (20×0.5) to the numerator.

We estimate three specifications: (1) own ranking alone (as above); (2) the outcome-matched field-composition variable alone; and (3) own ranking plus field composition. All three specifications exclude the 3 ego-rows whose own nationality is represented by exactly one player in the ego-row sample, so that a single lone-nationality observation cannot by itself drive any coefficient; this leaves N = 3,887 (from 3,890) throughout. The specification is deliberately parsimonious: the purpose is not to explain partner selection comprehensively, but to test whether the resulting cultural similarity of the partner is correlated with a player's own ability within a given tournament-year.

Table 9: Culture Similarity of the Partnership and Own Ability

Outcome	Variable	(1)	(2)	(3)
Same nationality (AME)	own_rank	0.000266***		0.000104*
		(0.000060)		(0.000059)
	composition		3.826***	3.734***
			(0.272174)	(0.290074)
Same language (AME)	own_rank	0.000189***		0.000152**
		(0.000065)		(0.000059)
	composition		1.270***	1.257***
			(0.094932)	(0.096450)
Ling. proximity (OLS coef.)	own_rank	0.000114**		0.000101***
		(0.000046)		(0.000039)
	composition		1.370***	1.366***
			(0.082830)	(0.083325)
N (focal player)		3,887	3,887	3,887

*** p<0.01, ** p<0.05, * p<0.10. Standard errors in parentheses, clustered by partnership. All specifications include tournament×year FE and exclude focal player whose own nationality has exactly 1 observation in the sample (3 observations dropped). Field cultural composition is (# other entrants sharing the same nationality, language as the focal player)/(N −1), and the analogous mean pairwise value for same linguistic proximity, computed against every other player entering that tournament-year.

The results are reported in Table 9. In Column (1) , the coefficient on own ranking is positive and statistically significant for all three measures: 0.000266 for same nationality (p¡0.001), 0.000189 for same language (p=0.004), and 0.000114 for linguistic proximity (p=0.014; OLS coefficient rather than AME, since this measure is continuous). Because a

lower doubles-ranking number indicates a stronger player, the positive coefficient means that worse-ranked players are, if anything, more likely to have culturally similar partners. Field cultural composition alone (Spec 2) is also a strong, highly significant predictor of all three outcomes, as expected, since a player drawn from a tournament field with more compatriots or linguistically similar entrants is mechanically more likely to end up with a culturally similar partner. Adding both own ranking and field composition together (Spec 3) shrinks the own-ranking coefficient substantially for same nationality (from 0.000266 to 0.000104) and only modestly for same language and linguistic proximity, but leaves it positive and statistically significant for all three outcomes ($p<0.05$ for same language and linguistic proximity; $p=0.078$ for same nationality), while the composition coefficient stays close to its Spec 2 magnitude. Own ranking therefore predicts partner similarity *in addition to*, not merely *because of*, the cultural composition of the field a player happens to be drawn from.

This result is the opposite of what would be expected if the performance association were primarily generated by stronger players having greater partner choice and systematically selecting culturally similar teammates. Together with the bracket comparisons, the evidence therefore provides no support for the specific selection mechanism in which stronger players disproportionately sort into culturally similar partnerships. It does not, however, eliminate other forms of endogenous partnership formation, and the performance estimates remain associations rather than causal effects.

Overall, the evidence in this section provides useful context for interpreting the main performance results. Cultural sorting is clearly present, but the data do not support the specific hypothesis that stronger players are disproportionately responsible for that sorting. At the same time, partnerships are formed through a mixture of deliberate choice and circumstances that can constrain partner selection. We therefore do not interpret the main estimates as causal effects of cultural similarity. Rather, the team-formation evidence helps assess the plausibility of one important alternative explanation for the observed association between cultural similarity and team performance.

Taken together, the partnership-formation evidence provides two useful qualifications to the performance analysis. First, cultural sorting in doubles is substantial: culturally similar partnerships are much more common than random matching would imply. Sec-

ond, the sorting does not appear to be disproportionately concentrated among higher-ranked players. These findings do not eliminate selection as a possible source of the performance association, but they make the specific explanation that stronger players systematically select culturally similar partners less consistent with the data.

6 Conclusion

In a world filled with global freelance platforms, decentralized autonomous organizations (DAOs), academic networks, and highly autonomous consultancies, individuals act as their own human resource managers.

Attention: tennis doubles are formed by individual players, not by a manager or an organization. Different implications if were assigned by managers/organizations like football. The findings also have implications for organizations in which employees have discretion over their collaborators. In such settings, team composition may emerge partly through endogenous cultural sorting rather than solely through managerial assignment. If cultural similarity facilitates coordination, such sorting may reflect an economically meaningful feature of decentralized team formation. At the same time, our results do not imply that managers should systematically favor culturally similar teams: the evidence is observational, and culturally homogeneous partnerships may also reflect selection into partnerships on unobserved characteristics.

While prior literature often evaluates team diversity through the lens of centralized, top-down managerial assignment, our findings offer vital insights for the expanding landscape of decentralized, self-organized collaborations. For the autonomous professional navigating global knowledge markets, we show that cultural homophily is not merely a psychological comfort, but a rational coordination subsidy that enhances performance under real-time strategic constraints. However, we identify a sharp boundary condition for individual strategy: the value of this cultural conduit is strictly dependent on the availability of strategic communication windows and the total pool of accumulated human capital. When tasks transition into hyper-velocity execution shocks governed by universal, protocol-driven signaling, the homophily premium evaporates. For organizational designers, this reveals that team performance can be dynamically managed not by forcing composition, but by structurally altering the temporal architecture of the

task environment.

Cultural diversity is increasingly a feature of how organizations assemble teams, but the organizational value of diversity depends not only on what individuals bring to a team but also on how effectively they can combine those resources. Our findings suggests that cultural similarity is associated with better team performance, but that this association is neither universal nor mechanical. Its relevance appears to depend on whether teammates have sufficient opportunity to communicate, develop shared understandings, and translate those understandings into coordinated action.

This perspective complements the recent finding that cultural homophily shapes collaboration even among highly skilled professionals in multinational football teams (Békés and Ottaviano, 2025). Their evidence shows that cultural similarity affects how teammates interact; our evidence suggests that such differences in interaction can also be reflected in collective performance. The distinction is important for management research. If cultural similarity matters partly because it lowers coordination costs, then its organizational consequences should depend on the structure of the task. Teams performing highly interdependent work, working under substantial time pressure, or relying on repeated interaction may be particularly sensitive to these costs. Conversely, organizations may be able to capture the informational benefits of cultural diversity when tasks allow greater time for communication, clearer division of responsibilities, or other mechanisms that reduce the need for implicit coordination.

The performance relevance of cultural similarity depends on whether the structure of the task allows communication and coordination advantages to accumulate.

The findings also suggest a more nuanced view of team composition. A natural managerial response to evidence of coordination costs would be to favor culturally similar employees when assembling teams. Our results do not justify such a prescription. Cultural similarity is strongly associated with partnership formation, but the evidence does not indicate that the strongest players are disproportionately responsible for this sorting. More importantly, our analysis is observational and cannot establish that cultural similarity itself causes superior performance. Cultural similarity may proxy for other dimensions of compatibility, partnership history, social networks, or unobserved characteristics that influence both team formation and performance. The evidence on

experience is consistent with a role for accumulated shared experience, but it does not identify the precise process through which communication advantages develop.

Cultural similarity may be more valuable when teams perform tasks requiring repeated interdependent coordination, communication, and adaptation. The value may be less visible when tasks are sufficiently short, tightly constrained, or dominated by small differences in performance. Managers should think about cultural composition jointly with task structure, rather than treating cultural similarity as inherently good or bad. Our evidence suggests that its performance relevance depends on whether team members have sufficient opportunity to communicate, adjust, and coordinate their actions over the course of a task.

While our empirical setting offers unique advantages in terms of observable effort, organic team formation, and clean performance metrics, several limitations remain that present fruitful avenues for future research. First, we emphasize that our empirical strategy identifies robust conditional associations rather than definitive causal effects. Although our models include comprehensive controls for individual and opponent capabilities, team experience, and tournament-by-year fixed effects—and our subsequent analysis rules out the alternative explanation that higher-ability actors systematically sort into homogeneous clusters—we cannot entirely eliminate the presence of unobservable, time-varying confounders. Factors such as unmeasured personal compatibility, real-time physical fatigue, or unobserved shared training histories could simultaneously influence both the decision to collaborate and the probability of match success. Future research utilizing true exogenous assignment of partners within high-velocity, real-time execution environments would be required to firmly establish a causal interpretation of the coordination channels we document. Second, while our findings demonstrate that the positive association between cultural similarity and performance completely disappears under the acute time shocks of tiebreaks, our data structure does not allow us to directly record the exact verbal or non-verbal content of the communication occurring during the 90-second changeovers versus between points. We infer that this null finding is driven by a structural shift from explicit strategic troubleshooting to standardized, protocol-driven tactical signaling. Future organizational studies could build upon our framework by capturing granular audio or visual tracking of self-organized

teams in action, explicitly measuring how the linguistic complexity of real-time pivots changes as communication windows shrink. Finally, our sample is exclusively restricted to professional men's teams operating in a hyper-competitive sports environment. Examining whether these conditional associations and temporal boundaries hold in other decentralized, self-organized knowledge settings—such as cross-border high-frequency trading duos or remote open-source software crisis units—would greatly enhance the generalizability of our findings.

These limitations point toward an important agenda for future research. The central challenge is to move from observing cultural similarity to identifying the coordination process through which similarity affects team production. Experimental or quasi-experimental variation in team composition could help separate cultural compatibility from selection into partnerships. Richer longitudinal data could distinguish shared experience with a particular teammate from general professional experience and examine whether coordination routines develop through repeated interaction. Direct measures of communication—such as verbal exchanges, shared decision-making, or behavioral measures of anticipation—could provide an even more direct test of whether cultural and linguistic similarity operate through communication costs and shared mental models.

More broadly, the paper suggests that the consequences of diversity should be understood as a property of the interaction between team composition and task design, rather than as an inherent feature of diversity itself. Cultural differences may provide organizations with access to heterogeneous knowledge, perspectives, and networks, while simultaneously increasing the effort required to integrate those resources. Whether the balance is favorable depends on the extent to which an organization's production process relies on rapid, repeated, and interdependent coordination. Understanding this interaction is important for managers designing teams, but also for economists and organizational scholars seeking to explain when diversity becomes an asset, when it becomes a coordination challenge, and how organizations can reduce the costs of bringing different people together.

A Alternative Measures of Linguistic Proximity

The main text measures linguistic proximity using the continuous $[0, 1]$ index of Melitz and Toubal (2014), built from Ethnologue's language-tree classification (the same measure used by Békés and Ottaviano (2025)). This Appendix reports every main-text regression table (3, 4, 5, 6, 6b) re-estimated with three alternative constructions of linguistic proximity, one at a time, in place of the main-text measure: (i) the original binary CEPII (Conte et al. 2022) *comlang_ethno* indicator; and two further continuous, independently-constructed robustness measures from Pellegrino et al. (2024), (ii) an inverted tree-distance index and (iii) a cognate/lexical-based proximity index. Both Pellegrino et al. (2024) measures are built on the same underlying idea as the main-text measure: languages related through a common ancestor are scored as linguistically closer; but are built from independently-constructed data with substantially broader language coverage. The tree-based measure computes, for any two languages, the ratio of the shortest path between them in Ethnologue's full language-family tree (6,737 languages) to the length of the path connecting them via the tree's root, giving a continuous distance on $[0, 1]$ that we invert into a proximity measure; as with the main-text measure, country-pair values are population-weighted averages over each country's spoken languages. The cognate-based measure instead scores two languages by the share of a core vocabulary list (584 concepts, drawn from the intersection of the Intercontinental Dictionary Series, Princeton WordNet, and the CogNet cognate database covering 338 languages) for which the two languages have cognate words (i.e., words descending from the same etymological root) again aggregated to the country-pair level by population share. All three achieve complete coverage on our sample. Every table below uses the identical sample, controls, and fixed effects as its main-text counterpart.

Table 10: Appendix A3 – Match Win, Linguistic Proximity Robustness (cf. Table 3)

	(1) Binary (comlang_ethno)	(2) PSW tree	(3) PSW cognate
Linguistic proximity	0.0380**	0.0648***	0.0719***
	(0.0157)	(0.0231)	(0.0231)
Own avg. doubles ranking	−0.0007***	−0.0007***	−0.0007***
	(0.0001)	(0.0001)	(0.0001)
Opponent avg. doubles ranking	0.0007***	0.0007***	0.0007***
	(0.0001)	(0.0001)	(0.0001)
Top-100 singles player (0,1)	−0.0026	−0.0038	−0.0034
	(0.0315)	(0.0314)	(0.0315)
Team avg. professional years	0.0042**	0.0046***	0.0047***
	(0.0017)	(0.0017)	(0.0017)
Team avg. professional years^2	−0.0007***	−0.0008***	−0.0008***
	(0.0002)	(0.0002)	(0.0002)
Tournament × year FE	Yes	Yes	Yes
Match round FE	Yes	Yes	Yes
Pseudo-R^2	0.096	0.097	0.097
N	3,728	3,728	3,728

*** p<0.01, ** p<0.05, * p<0.10. Standard errors in parentheses, clustered by match. Column (1) reproduces the pre-2026-09-10 main-text Table 3 column (3) exactly. Columns (2)-(3) AME reported per full-scale (0→1) change.

Table 11: Appendix A4 – Tiebreak Win, Linguistic Proximity Robustness (cf. Table 5)

	(1) Binary (comlang_ethno)	(2) PSW tree	(3) PSW cognate
Linguistic proximity	0.0031	0.0592*	0.0572*
	(0.0217)	(0.0312)	(0.0321)
Own avg. doubles ranking	−0.0003***	−0.0003***	−0.0003***
	(0.0001)	(0.0001)	(0.0001)
Opponent avg. doubles ranking	0.0003***	0.0003***	0.0003***
	(0.0001)	(0.0001)	(0.0001)
Top-100 singles player (0,1)	−0.0645	−0.0684	−0.0677
	(0.0427)	(0.0428)	(0.0428)
Team avg. professional years	0.0022	0.0029	0.0029
	(0.0021)	(0.0022)	(0.0022)
Team avg. professional years^2	0.0001	0.0000	0.0000
	(0.0003)	(0.0003)	(0.0003)
Tournament × year FE	Yes	Yes	Yes
Round FE	Yes	Yes	Yes
Pseudo-R^2	0.013	0.014	0.014
N	2,354	2,354	2,354

*** p<0.01, ** p<0.05, * p<0.10. Standard errors in parentheses, clustered by match. Column (1) reproduces the pre-2026-09-10 main-text Table 5 column (3) exactly.

Table 12: Appendix A5 – Culture × Surface, Linguistic Proximity Robustness (cf. Table 4)

	(1) Binary		(2) PSW tree		(3) PSW cognate	
	(a) grass	(b) clay	(a) grass	(b) clay	(a) grass	(b) clay
Cultural similarity (C)	0.0369**	0.0365**	0.0687***	0.0610**	0.0751***	0.0708***
	(0.0176)	(0.0181)	(0.0256)	(0.0268)	(0.0257)	(0.0268)
Surface	−0.0003	−0.0032	0.0148	−0.0083	0.0133	−0.0018
	(0.0209)	(0.0195)	(0.0337)	(0.0308)	(0.0340)	(0.0314)
C × Surface	0.0045	0.0051	−0.0197	0.0124	−0.0171	0.0021
	(0.0364)	(0.0344)	(0.0539)	(0.0492)	(0.0536)	(0.0495)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Round FE	Yes	Yes	Yes	Yes	Yes	Yes
Tournament × year FE	No	No	No	No	No	No
Pseudo-R^2	0.096	0.096	0.097	0.097	0.097	0.097
N	3,728		3,728		3,728	

*** p<0.01, ** p<0.05, * p<0.10. Standard errors in parentheses, clustered by match. Column (1) reproduces the pre-2026-09-10 main-text Table 4 column (3) exactly.

Table 13: Appendix A6 – Culture × Years Since Turning Pro, Linguistic Proximity Robustness (cf. Table 6)

	(1) Binary	(2) PSW tree	(3) PSW cognate
Cultural similarity (C)	0.0353**	0.0642***	0.0705***
	(0.0156)	(0.0230)	(0.0231)
C × exp_mean_dm	0.0069**	0.0048	0.0053
	(0.0032)	(0.0047)	(0.0047)
exp_mean_dm (main)	0.0000	–	–
	(0.0025)		
Controls	Yes	Yes	Yes
Tournament × year FE	Yes	Yes	Yes
Round FE	Yes	Yes	Yes
Pseudo-R^2	0.097	0.097	0.097
N	3,728	3,728	3,728

*** p<0.01, ** p<0.05, * p<0.10. Standard errors in parentheses, clustered by match. Column (1) reproduces the pre-2026-09-10 main-text Table 6 column (3) exactly and is the specification with a statistically significant culture × experience interaction referenced in Section 5. Columns (2)-(3)'s main exp_mean_dm control coefficient was not separately recorded in this run; the interaction and C-main rows are unaffected.

Table 14: Appendix A6b – Culture × Within-Team Experience Gap, Linguistic Proximity Robustness (cf. Table 7)

	(1) Binary	(2) PSW tree	(3) PSW cognate
Culture (C) – mean gap	0.0370**	0.0620***	0.0690***
	(0.0157)	(0.0231)	(0.0231)
C × exp_gap_dm	0.0038	0.0052	0.0052
	(0.0032)	(0.0048)	(0.0047)
exp_gap_dm (main)	−0.0046*	−0.0055	−0.0055
	(0.0025)	(0.0034)	(0.0033)
Controls (Table 3 set)	Yes	Yes	Yes
Tournament × year FE	Yes	Yes	Yes
Round FE	Yes	Yes	Yes
Pseudo-R^2	0.097	0.097	0.098
N	3,728	3,728	3,728

*** p<0.01, ** p<0.05, * p<0.10. Standard errors in parentheses, clustered by match. Column (1) reproduces the pre-2026-09-10 main-text Table 7 column (3) exactly.

Across every table, the binary CEPII measure and the two PSW2024 continuous measures agree qualitatively with the main-text linguistic-proximity results, with one exception noted in Section 5: the binary measure's culture × experience interaction (Appendix Table 13, column 1) is statistically significant, while the main-text linguistic-proximity interaction and both PSW2024 robustness interactions are not.

B Robustness to Double-Counting: One Randomly-Selected Team per Match

Table 3 contributes two observations per match (winner team and loser team), which are mechanically related: a win for one team is a loss for the other. Clustering standard errors by match already accounts for this non-independence, but as a further check we re-estimate the Table 3 specification keeping exactly one, randomly selected team per match (N = 1,864), which removes the mechanical win/loss complementarity entirely.

Table 15: Appendix B1 – Match Win, One Randomly-Selected Team per Match (cf. Table 3)

	(1) Same Nationality	(2) Same Language	(3) Lang. Proximity
Cultural similarity	0.0466	0.0477	0.0656
	$(0.0220)^{5/10}$	$(0.0207)^{6/10}$	$(0.0242)^{7/10}$
Own avg. doubles ranking	−0.0007	−0.0007	−0.0007
	$(0.0001)^{10/10}$	$(0.0001)^{10/10}$	$(0.0001)^{10/10}$
Opponent avg. doubles ranking	0.0008	0.0008	0.0008
	$(0.0001)^{10/10}$	$(0.0001)^{10/10}$	$(0.0001)^{10/10}$
Top-100 singles player (0,1)	−0.0099	−0.0078	−0.0080
	$(0.0359)^{0/10}$	$(0.0356)^{0/10}$	$(0.0354)^{0/10}$
Team avg. professional years	0.0050	0.0047	0.0049
	$(0.0019)^{7/10}$	$(0.0018)^{7/10}$	$(0.0019)^{7/10}$
Team avg. professional years^2	−0.0008	−0.0008	−0.0008
	$(0.0003)^{7/10}$	$(0.0003)^{7/10}$	$(0.0003)^{7/10}$
Tournament × year FE	Yes	Yes	Yes
Match round FE	Yes	Yes	Yes
Pseudo-R^2 (mean)	0.112	0.113	0.113
N (per draw)	1,864	1,864	1,864

Each draw keeps exactly one team per match (coin flip, N = 1,864), estimated with the same controls and fixed effects (tournament × year, round) as Table 3, with robust (HC1) standard errors in place of match-clustering. Statistics computed across 10 independent random seeds. Each coefficient is the mean AME across the 10 draws; the number in parentheses is the standard deviation of that AME *across draws*. The superscript after each parenthesis instead reports, for that coefficient in that column, how many of the 10 individual draws were significant at the 5% level *in that draw's own regression*.

The randomly-selected-team specification produces AMEs of similar magnitude to Table 3's main estimates for all three measures, with somewhat weaker and less stable significance, as expected given the sample is halved and no longer benefits from the efficiency gain of clustering two related observations per match.

C Disentangling Nationality from Language

In this sample every same-nationality partnership also shares an official language (no same-nationality/different-language team-observations exist). The two binary measures therefore define three mutually exclusive categories that partition the sample:

$$C_{ij} = \begin{cases} 0 & \text{different nationality, different language} \quad (N = 1{,}619;\ \text{raw win rate } 0.490) \\ 1 & \text{different nationality, same language} \quad (N = 575;\ \text{raw win rate } 0.557) \\ 2 & \text{same nationality, same language} \quad (N = 1{,}534;\ \text{raw win rate } 0.490) \end{cases}$$

We estimate

$$Y_{ijt} = \delta_1 \mathbf{1}(C_{ij} = 1) + \delta_2 \mathbf{1}(C_{ij} = 2) + X'_{ijt}\gamma + FE_t + \varepsilon_{ijt},$$

with category 0 as the reference group and the same controls, fixed effects, and clustering as Table 3. δ_1 measures the association with a shared language across nationalities, δ_2 the association with shared nationality and language, and $\delta_2 - \delta_1$ the additional association with same nationality conditional on a shared language.

Table 16: Appendix C1 – Match Win by Nationality–Language Category (cf. Table 3)

	Match win
δ_1: Different nationality & same language	0.0353
	(0.0227)
δ_2: Same nationality & same language	0.0449***
	(0.0173)
Own avg. doubles ranking	−0.0007***
	(0.0001)
Opponent avg. doubles ranking	0.0007***
	(0.0001)
Top-100 singles player (0,1)	−0.0036
	(0.0315)
Team avg. professional years	0.0043***
	(0.0017)
Team avg. professional years^2	−0.0008***
	(0.0002)
$\delta_2 - \delta_1$	0.0096
	(0.0237)
Tournament × year FE	Yes
Round FE	Yes
Pseudo-R^2	0.097
N	3,728

*** p<0.01, ** p<0.05, * p<0.10. Average marginal effects; standard errors in parentheses, clustered by match. Reference category: different nationality & different language. p-values: δ_1 0.119, δ_2 0.010, $\delta_2 - \delta_1$ 0.687.

Relative to fully mixed partnerships, same-nationality partnerships are 4.5 percentage points more likely to win ($p = 0.010$). Partnerships of different nationalities that share a language are 3.5 percentage points more likely to win; this is about 80% of the same-nationality association, but it is not significant ($p = 0.119$), reflecting the small size of this group ($N = 575$). The additional association with same nationality, conditional on a shared language ($\delta_2 - \delta_1$), is small (0.010) and far from significant ($p = 0.687$).

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